

Reinforcement Learning

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Lecture 7: Imitation Learning

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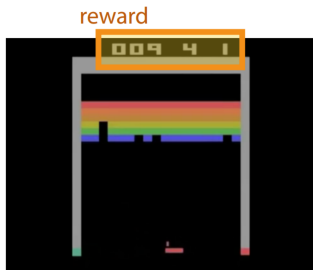


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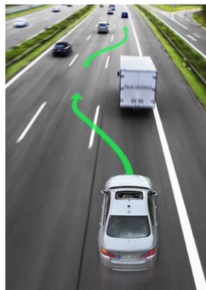
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Learning from demonstrations (LfD)

- Motivation:**
- In RL, the reward function is known and we maximize the cumulative reward.
 - The reward functions are often manually designed to define the task.
 - Can we instead learn a policy by capitalizing an expert's behavior?

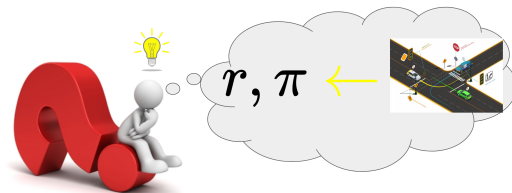


(a)



(b)

Learning from demonstrations (LfD) (cont'd)



Real world problems:

- The reward function is unknown or is difficult to be designed.
- It is easier/more natural to use “demonstrations” by experts.

Imitation learning (IL) vs inverse reinforcement learning (IRL)

- Setting:

- ▶ Given expert demonstrations $\mathcal{D}_E = \{(s_i, a_i)\}_{i=1}^n$ or trajectories
- ▶ Reward signal is unobserved
- ▶ Transition model may be known or unknown

- Goals and approaches:

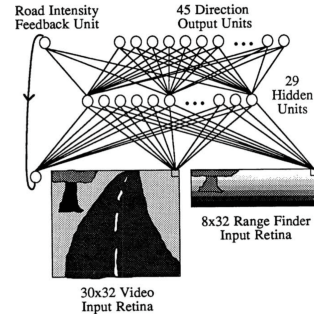
- ▶ Learn the expert's policy π_E directly: **imitation learning (IL)**
- ▶ Infer a reward function under which expert behavior is near-optimal: **inverse reinforcement learning (IRL)**

A historic application

- Imitation learning appeared early in autonomous driving, predating the formal IRL framework of [25].



(c)



(d)

Figure: One of the first imitation learning systems using neural networks.

- ALVINN: Autonomous Land Vehicle In a Neural Network, 1989 [30].

<https://www.youtube.com/watch?v=2KMAAmkz9go&t=205s>.

A modern application: RLHF

- Large language models: ChatGPT



<https://www.forbes.com>



- The last training step is based on Reinforcement Learning from Human Feedback (RLHF) (see [27]).
- A recent work [47] shows a close connection between IRL and RLHF.

More imitation learning applications

- Simulated highway driving [2]
- Helicopter acrobatics [1]
- Urban navigation [48]
- Human goal inference [23]
- Object manipulation [36, 11]



(a)



(b)

Figure: Helicopter model and example aerobatic maneuver.

Source: [8]

Taxonomy of Learning from Demonstrations

Method	Reward learning	Access to environment	Interactive demonstrations	Pre-collected demonstrations
Behavioral Cloning	NO	NO	NO	YES
Online IL	NO	YES	YES	MAYBE
Inverse RL	YES	YES	NO	YES
Adversarial IL	MAYBE	YES	NO	YES
Non-adversarial IL	MAYBE	YES	NO	YES

- Remarks:**
- BC avoids interaction with the environment, but can suffer from cascading errors.
 - Online IL helps with the cascading errors but requires (expensive) expert queries.
 - IRL explains the expert's behavior but has poor sample complexity and scalability.
 - Adversarial IL avoids solving RL repeatedly but is unstable due to adversarial training.
 - Non-adversarial IL is often more stable, but its theory is less developed.

Offline IL: Behavioral cloning

- We observe demonstrations from an expert policy π_E .
- Input: offline expert data $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^n$, with $a_i \sim \pi_E(\cdot|s_i)$.

Offline IL: Behavioral cloning

- We observe demonstrations from an expert policy π_E .
- Input: offline expert data $\mathcal{D} = \{(s_i, a_i)\}_{i=1}^n$, with $a_i \sim \pi_E(\cdot|s_i)$.
- **Idea:** Learn π_E directly from state-action pairs using supervised learning.

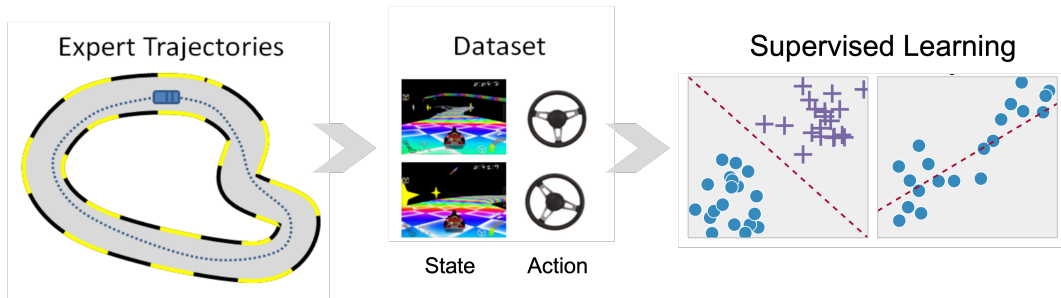


Figure: Source: <https://smartlabai.medium.com/a-brief-overview-of-imitation-learning-8a8a75c44a9c>

Behavioral cloning as supervised learning

Maximum Likelihood Estimation (MLE)

For a stochastic policy class Π , behavioral cloning can be written as maximum likelihood:

$$\hat{\pi}_{\text{MLE}} = \operatorname{argmax}_{\pi \in \Pi} \sum_{(s,a) \in \mathcal{D}} \log \pi(a|s). \quad (1)$$

Risk Minimization

Alternatively, following [3], we can try to minimize a loss between our parameterized policy π_θ and the expert policy π_E as

$$\min_{\theta} \mathbb{E}_{s \sim \lambda_{\mu}^{\pi_E}} \left[\ell \left(\pi_{\theta}(\cdot|s), \pi_E(\cdot|s) \right) \right], \quad (2)$$

where $\lambda_{\mu}^{\pi_E}$ is the state visitation distribution under policy π_E and ℓ is a loss function. Typically, the loss is cross-entropy or KL divergence.

Behavioral cloning: error compounding guarantee

- The following guarantee is from [3].

Theorem (Behavior Cloning)

Let Π be a discrete and realizable policy class, i.e., $\pi_E \in \Pi$, and define $J(\pi) = \langle \mu, V^\pi \rangle$. With probability at least $1 - \delta$, MLE behavioral cloning returns a policy that obeys

$$J(\pi_E) - J(\hat{\pi}_{\text{MLE}}) = \langle \mu, V^{\pi_E} - V^{\hat{\pi}_{\text{MLE}}} \rangle \leq \mathcal{O} \left(\frac{1}{(1-\gamma)^2} \sqrt{\frac{\log(|\Pi|/\delta)}{|\mathcal{D}|}} \right),$$

where $|\Pi|$ is the size of the policy class, and $|\mathcal{D}|$ is the length of the provided dataset.

- Idea:**
- BC learns from states visited by the expert policy π_E .
 - When deployed, the learned policy $\hat{\pi}_{\text{MLE}}$ may visit different states.
 - Small supervised errors can therefore compound into a larger control error.
 - The factor $\frac{1}{(1-\gamma)^2}$ is the signature of this error compounding.

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where $|\Pi|$ is the size of the policy class, and $|\mathcal{D}|$ is the length of the provided dataset.

Remarks: ◦ BC only ensures the learned policy $\hat{\pi}_{\text{MLE}}$ is close to π_E under the support of distribution $\lambda_\mu^{\pi_E}$.

◦ The term $\sqrt{\frac{\log(|\Pi|/\delta)}{|\mathcal{D}|}}$ reflects the error $\hat{\pi}_{\text{MLE}}$ and π_E under the distribution $\lambda_\mu^{\pi_E}$.

◦ The term $\frac{1}{(1-\gamma)^2}$ reflects the cascading errors when performing with respect to the policy $\hat{\pi}_{\text{MLE}}$.

◦ The bound improves to $\mathcal{O} \left(\frac{1}{(1-\gamma)} \sqrt{\frac{\log(|\Pi|/\delta)}{|\mathcal{D}|}} \right)$ in the finite horizon [12].

◦ The term $\frac{1}{(1-\gamma)^2}$ can be improved to $\frac{1}{1-\gamma}$ when the transition model is known [3].

Behavioral cloning: Strengths and failure mode

- Strengths

- Easy to train: reduces IL to supervised learning.
- Does not require environment interaction or reward design.
- Can work well when expert data covers the states encountered at test time.

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- Does not require environment interaction or reward design.
- Can work well when expert data covers the states encountered at test time.

- Failure mode

- The learned policy is trained on expert-visited states.
- Mistakes can move the agent off the expert distribution.
- Off-distribution states require recovery behavior that may be absent from the data.
- Ross et al. [34] highlight the following quote from Pomerleau.

Quote from Pomerleau

When driving for itself, the network (ALVINN) may occasionally stray from the center of road and so must be prepared to recover by steering the vehicle back to the center of the road.

Why BC Is Not Just Supervised Learning

- BC trains on states sampled from the expert visitation distribution $\lambda_{\mu}^{\pi_E}$:

$$\min_{\theta} \mathbb{E}_{s \sim \lambda_{\mu}^{\pi_E}} \left[\ell \left(\pi_{\theta}(\cdot|s), \pi_E(\cdot|s) \right) \right].$$

- However, when we act in the environment with π_{θ} , states are sampled according to $\lambda_{\mu}^{\pi_{\theta}}$.
- Hence, ideally we would like to minimize

$$\min_{\theta} \mathbb{E}_{s \sim \lambda_{\mu}^{\pi_{\theta}}} \left[\ell \left(\pi_{\theta}(\cdot|s), \pi_E(\cdot|s) \right) \right].$$

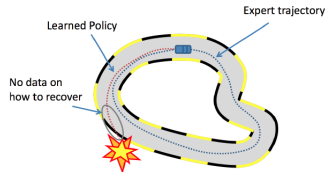
- This feedback loop is absent in ordinary supervised learning:
 - ▶ prediction errors do not change the future test inputs.

From BC to interactive IL

- BC trains only on expert-visited states.
- Deployment reveals the states induced by the learned policy.
- Interactive IL queries the expert on those states.
- Interactive IL then adds them to the training set.

- **Key idea:** collect labels where the learner actually goes.

Figure: <https://smartlabai.medium.com/a-brief-overview-of-imitation-learning-8a8a75c44a9c>



Interactive IL: querying the expert

- Mitigates cascading errors by collecting expert labels on learner-visited states.
- We assume query access to the expert π_E on states sampled from $\lambda_{\mu}^{\pi_{\theta}}$.
- **Idea:** update the policy online as new expert-labeled states arrive.

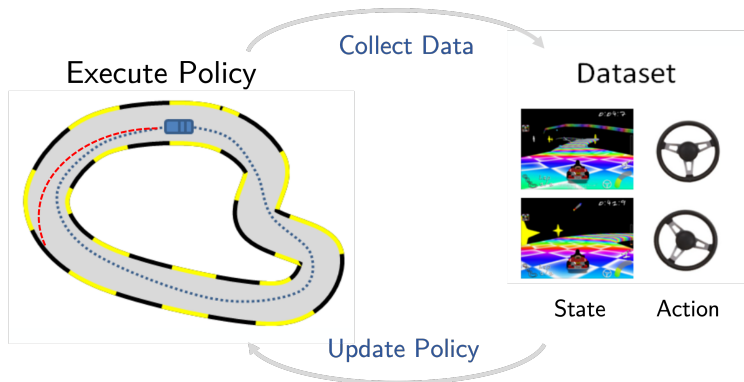


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Interactive IL algorithms: DAgger and SMILe

- [Dataset Aggregation](#) (DAgger) [33]: train on the union of all expert-labeled learner states.
- [Policy Aggregation](#) (e.g., SMILe [34]): update by mixing the previous policy with a newly trained policy.

Interactive IL algorithms: DAgger and SMILe

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Interactive imitation learning

Initialize π_1

for each iteration $t = 1, \dots, T$ **do**

 Generate trajectories τ following π_t

 Collect new data $\mathcal{D}_t = \{(s, \pi_E(s)) | s \in \tau\}$ based on expert's feedback

Data Aggregation: run behavioral cloning with $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2 \cup \dots \cup \mathcal{D}_t$ and obtain π_{t+1}

Policy Aggregation: run behavioral cloning with $\mathcal{D}_t$ and obtain $\hat{\pi}_t$, set $\pi_{t+1} = \beta \hat{\pi}_t + (1 - \beta) \pi_t$

end for

- Remark:**
- In the dataset $\mathcal{D}_t$ the states are sampled according to λ^{π_t} .
 - However, the actions are sampled from π_E . We need to assume that the expert is interactive.

Interactive IL as online decisions

- At round t , the learner's decision is a policy $\pi_t \in \Pi$.
- Rolling out π_t induces states sampled from $\lambda_\mu^{\pi_t}$.
- The expert labels these states, producing data $\mathcal{D}_t = \{(s, \pi_E(s)) : s \sim \lambda_\mu^{\pi_t}\}$.
- This round defines a loss for any candidate policy π :

$$\ell_t(\pi) = \mathbb{E}_{s \sim \lambda_\mu^{\pi_t}} \left[\ell(\pi(\cdot|s), \pi_E(\cdot|s)) \right].$$

- Key point: the learner's policy π_t determines the loss ℓ_t by determining which states are visited.

No-regret online learning

- Online learning studies repeated decisions with losses revealed over time.

Online learning protocol

- At round t , the learner chooses a decision $\mathbf{x}_t \in X$.
- Then a loss function $\ell_t(\cdot) : X \rightarrow \mathbb{R}$ is revealed.
- The learner suffers loss $\ell_t(\mathbf{x}_t)$.
- Regret compares the learner to the best fixed decision in hindsight:

$$\mathcal{R}_T = \sum_{t=1}^T \ell_t(\mathbf{x}_t) - \min_{\mathbf{x} \in X} \sum_{t=1}^T \ell_t(\mathbf{x}).$$

- A no-regret algorithm satisfies $\mathcal{R}_T = o(T)$.

Follow-the-Leader

- Follow-the-Leader (FTL) chooses the decision that would have performed best on all past losses:

$$\mathbf{x}_{t+1} = \arg \min_{\mathbf{x} \in X} \sum_{i=1}^t \ell_i(\mathbf{x}).$$

- In words: after observing rounds $1, \dots, t$, pick the best fixed decision for those rounds.
- FTL is not always no-regret, but it has no-regret guarantees under standard stability/convexity assumptions.

DAGger is Follow-the-Leader

- DAGger replaces the population loss by the empirical loss on expert-labeled learner states:

$$\hat{\ell}_t(\pi) = \sum_{s \in \mathcal{D}_t} \ell(\pi(\cdot|s), \pi_E(\cdot|s)).$$

- DAGger aggregates all data collected so far and retrains:

$$\pi_{t+1} = \arg \min_{\pi \in \Pi} \sum_{i=1}^t \hat{\ell}_i(\pi).$$

- Thus, DAGger is FTL with decision $\mathbf{x} = \pi$ and losses $\hat{\ell}_t$ induced by learner-visited states.

Why no-regret matters for IL

- The reduction connects cumulative control error to online-learning regret:

$$\sum_{t=1}^T \left(J(\pi_E) - J(\pi_t) \right) \lesssim \frac{\max_{s,a} |Q^{\pi_E}(s,a)|}{1-\gamma} \left(\mathcal{O}(\sqrt{T}) + \mathcal{R}_T \right).$$

- The $\mathcal{O}(\sqrt{T})$ term comes from estimating learner-induced losses from sampled trajectories.
- The $\mathcal{R}_T$ term is controlled by the online learning algorithm.
- If regret is sublinear, the average performance gap is controlled with only one horizon factor.
- Recall that $J(\pi) = \langle \mu, V^\pi \rangle$, where μ is the initial state distribution.

The reduction

- We bound cumulative performance loss by imitation loss on learner-induced states.

- Define $d_t(\pi) = \mathbb{E}_{s \sim \lambda_\mu^{\pi_t}} \left[\|\pi_E(\cdot|s) - \pi(\cdot|s)\|_1 \right]$, $\hat{d}_t(\pi) = \widehat{\mathbb{E}}_{s \in \mathcal{D}_t} \left[\|\pi_E(\cdot|s) - \pi(\cdot|s)\|_1 \right]$.

Proof:

$$\sum_{t=1}^T \langle \mu, V^{\pi_E} - V^{\pi_t} \rangle = \sum_{t=1}^T \frac{1}{1-\gamma} \mathbb{E}_{s \sim \lambda_\mu^{\pi_t}} \left[\langle Q^{\pi_E}(s, \cdot), \pi_E(\cdot|s) - \pi_t(\cdot|s) \rangle \right] \quad (\text{PDL})$$

$$\begin{aligned} &\leq \frac{\max_{s,a} |Q^{\pi_E}(s, a)|}{1-\gamma} \sum_{t=1}^T d_t(\pi_t) \\ &= \frac{\max_{s,a} |Q^{\pi_E}(s, a)|}{1-\gamma} \left(\sum_{t=1}^T (d_t(\pi_t) - \hat{d}_t(\pi_t)) + \sum_{t=1}^T \hat{d}_t(\pi_t) \right) \\ &\leq \frac{\max_{s,a} |Q^{\pi_E}(s, a)|}{1-\gamma} \left(\mathcal{O}(\sqrt{T}) + \mathcal{R}_T \right). \end{aligned}$$

- The term $\sum_{t=1}^T (d_t(\pi_t) - \hat{d}_t(\pi_t)) = \mathcal{O}(\sqrt{T})$ follows from Azuma-Hoeffding concentration.
- The term $\sum_{t=1}^T \hat{d}_t(\pi_t)$ is controlled by online-learning regret.
- DAgger uses FTL; SMILe is better viewed through a conservative policy-mixing lens.

Dagger: Consequence and evidence

- The no-regret reduction improves the error inflation factor from

$$\mathcal{O}\left(\frac{1}{(1-\gamma)^2}\right) \quad \text{to} \quad \mathcal{O}\left(\frac{\max_{s,a} |Q^{\pi_E}(s,a)|}{1-\gamma}\right)$$

for interactive IL methods such as DAgger [3].

- Intuition: DAgger trains on the states induced by the learner, so recovery behavior enters the supervised data.

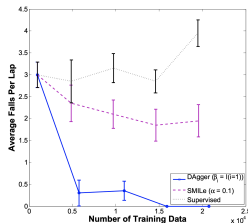


Figure: 3D racing car [33]

SMILe: The viewpoint of the original paper

- Ross and Bagnell connect SMILe to CPI and SEARN [34, 21, 10].
- SMILe updates by conservative policy mixing.

$$\pi_{t+1} = (1 - \beta_t)\pi_t + \beta_t\hat{\pi}_t.$$

- Here $\hat{\pi}_t$ is trained on learner-induced data $\mathcal{D}_t$.
- This is policy aggregation, not a no-regret reduction.

SMILe: A Frank-Wolfe lens

- Later work interprets CPI as Frank-Wolfe [41].
- This suggests a FW view for SMILe on $X = \text{conv}(\Pi)$.

$$\hat{\ell}_t(\pi) = \sum_{s \in \mathcal{D}_t} \ell(\pi(\cdot|s), \pi_E(\cdot|s)).$$

$$s_t \in \arg \min_{\pi \in \Pi} \langle \nabla \hat{\ell}_t(\pi_t), \pi \rangle.$$

$$\pi_{t+1} = (1 - \eta_t)\pi_t + \eta_t s_t.$$

- The linear oracle uses the current dataset $\mathcal{D}_t$.
- The classifier $\hat{\pi}_t$ plays the role of this direction.

- Remarks:**
- Classical FW rates need variance control in this setting.
 - Modern fixes include SPIDER-FW [45].
 - Norm-constrained LMOs also appear in Scion [29].

Imitation learning from reward features only

- The methods seen so far require observing expert actions or querying the expert.
- In some applications, we only observe trajectories or state features, not the actions that produced them.

Examples: ○ Driving videos show vehicle motion, but not the driver's control inputs.
○ Cooking videos show states and outcomes, but not low-level robot actions [13].



Figure: Robot learning to cook from videos.

- Next, we study IL methods that use reward features and assume $r_w(s, a) = \langle w, \phi(s, a) \rangle$.

Feature expectation matching

- Assume linear rewards $r_w(s, a) = w^\top \phi(s, a)$ with features $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$.
- Define the feature expectation of a policy:

$$\rho_\phi(\pi) := \mathbb{E}_{(s,a) \sim \lambda_\mu^\pi} [\phi(s, a)].$$

- Matching feature expectations controls performance for linear rewards:

$$\begin{aligned} J_w(\pi_E) - J_w(\pi) &= \langle \mu, V_w^{\pi_E} - V_w^\pi \rangle \\ &\leq \frac{1}{1-\gamma} w^\top (\rho_\phi(\pi_E) - \rho_\phi(\pi)) \\ &\leq \frac{\|w\|_2}{1-\gamma} \|\rho_\phi(\pi_E) - \rho_\phi(\pi)\|_2. \end{aligned}$$

- Therefore, it suffices to solve

$$\min_{\pi} \|\rho_\phi(\pi) - \rho_\phi(\pi_E)\|_2^2. \quad (3)$$

Apprenticeship learning as minimax matching

- Feature matching is a special case of a broader minimax principle.
- Assume the true reward belongs to a reward class $\mathcal{R}$.

Apprenticeship learning formalism

$$\min_{\pi} \max_{r \in \mathcal{R}} J_r(\pi_E) - J_r(\pi) = \min_{\pi} \max_{r \in \mathcal{R}} \langle \lambda_{\mu}^{\pi_E} - \lambda_{\mu}^{\pi}, r \rangle. \quad (4)$$

- Remarks:**
- If $\mathcal{R} = \{w^{\top} \phi : \|w\|_2 \leq 1\}$, then $\max_{r \in \mathcal{R}} \langle \lambda_{\mu}^{\pi_E} - \lambda_{\mu}^{\pi}, r \rangle = \|\rho_{\phi}(\pi_E) - \rho_{\phi}(\pi)\|_2$,
so (3) is the Euclidean feature-matching case.
 - For general $\mathcal{R}$, the minimax form matches occupancies via the distance $\max_{r \in \mathcal{R}} \langle \lambda_{\mu}^{\pi_E} - \lambda_{\mu}^{\pi}, r \rangle$.
 - This is an integral probability metric [24]; different reward classes induce different distances.

Maximum entropy IRL

- MaxEnt IRL regularizes feature matching by preferring high-entropy policies [48].

Entropy-regularized saddle problem

Let λ_μ^π be the state-action occupancy measure of policy π . Consider the following problem:

$$\min_w \max_{\pi \in \Pi} w^\top \left(\mathbb{E}_{(s,a) \sim \lambda_\mu^\pi} [\phi(s,a)] - \mathbb{E}_{(s,a) \sim \lambda_\mu^{\pi_E}} [\phi(s,a)] \right) + \alpha \mathbb{E}_{(s,a) \sim \lambda_\mu^\pi} [-\log \pi(a|s)].$$

- Remarks:**
- Game-theoretic perspective: zero-sum game between the reward and the policy.
 - For fixed w , the inner problem is entropy-regularized RL.
 - The outer minimization updates w until the induced policy matches expert feature expectations.

Solving the MaxEnt saddle problem

- Define the outer objective

$$f(w) = \max_{\pi \in \Pi} w^\top \left(\rho_\phi(\pi) - \rho_\phi(\pi_E) \right) + \alpha \mathbb{E}_{(s,a) \sim \lambda_\mu^\pi} [-\log \pi(a|s)].$$

- For fixed w , evaluating $f(w)$ requires entropy-regularized RL:

$$\pi_w^* \in \arg \max_{\pi \in \Pi} \mathbb{E}_{(s,a) \sim \lambda_\mu^\pi} \left[w^\top \phi(s, a) - \alpha \log \pi(a|s) \right].$$

- By Danskin's theorem [9],

$$\nabla f(w) = \rho_\phi(\pi_w^*) - \rho_\phi(\pi_E).$$

Algorithm:

- Solve entropy-regularized RL for π_w^* .
- Update $w \leftarrow w - \eta \left(\rho_\phi(\pi_w^*) - \rho_\phi(\pi_E) \right)$.

Remark:

- The inner RL step is expensive and typically needs transition access or many samples.

Avoiding RL in the inner loop via linear programming

- MaxEnt IRL requires solving an RL problem for each reward update.
- We now optimize directly over occupancy measures to avoid this inner RL loop.
- Let $\mathcal{R}$ be a class of reward functions. The reward class tries to separate learner and expert occupancies.

LP for imitation learning

$$\max_{\lambda} \min_{r \in \mathcal{R}} \langle \lambda - \lambda_{\mu}^{\pi^E}, r \rangle \quad (5)$$

$$\text{s.t. } E^{\top} \lambda = \gamma P^{\top} \lambda + (1 - \gamma) \mu \quad (6)$$

Remarks:

- The decision variable is the occupancy measure $\lambda \in \mathbb{R}^{|S||\mathcal{A}|}$.
- The constraint is the discounted flow conservation equation.
- To improve scalability, [22] study the Lagrangian and use linear function approximation.

From the LP to a saddle-point problem

- We dualize the discounted flow conservation constraint.
- The dual variable is a value-like function $V : \mathcal{S} \rightarrow \mathbb{R}$.
- This gives a saddle-point problem over occupancy measure, reward, and dual variable.

Saddle point formulation for imitation learning

$$\max_{\lambda} \min_{r \in \mathcal{R}} \min_V \langle \lambda - \lambda_{\mu}^{\pi^E}, r \rangle + \langle V, -E^{\top} \lambda + \gamma P^{\top} \lambda + (1 - \gamma) \mu \rangle \quad (7)$$

Remarks:

- λ is the primal variable, the learner occupancy measure.
- r is the worst-case reward chosen from $\mathcal{R}$.
- V penalizes violations of the discounted flow constraint.
- This saddle form is the starting point for function approximation.

Making the saddle-point problem scalable

- The saddle-point formulation is still tabular and scales with $|\mathcal{S}|$ and $|\mathcal{A}|$.
- To scale up, restrict the occupancy, value, and cost functions to linear feature classes [22].

Reduced saddle-point formulation

$$\min_{\theta \in \Theta} \max_{\lambda \in \Lambda} \max_{w \in W} \langle \Phi\theta - \lambda_{\mu}^{\pi^E}, Cw \rangle + \langle \Psi\lambda, -E^{\top}\Phi\theta + \gamma P^{\top}\Phi\theta + (1 - \gamma)\mu \rangle \quad (8)$$

- Remarks:**
- Here $\mu_{\theta} = \Phi\theta$, $u_{\lambda} = \Psi\lambda$, and $c_w = Cw$.
 - The optimization now depends on parameter dimensions, not directly on $|\mathcal{S}|$ and $|\mathcal{A}|$.
 - The trade-off is a strong assumption on the chosen features [22, 6].

A stronger assumption: linear MDPs

- To go beyond generic feature approximation, we now assume the MDP itself has linear structure.

Linear MDP

The same feature map $\phi(s, a)$ controls both the reward and the transition model [20]:

$$\begin{aligned} r(s, a) &= \langle \phi(s, a), w \rangle, & w &\in \mathcal{W} := \{w \in \mathbb{R}^m : \|w\|_2 \leq 1\}, \\ P(s'|s, a) &= \langle \phi(s, a), g(s') \rangle, & g &: \mathcal{S} \rightarrow \mathbb{R}^m. \end{aligned}$$

Equivalently, in matrix form:

$$r = \Phi w, \quad P = \Phi M.$$

Remarks:

- This is stronger than generic function approximation.
- Examples include tabular MDPs and anchor-state-action models [43].
- In return, the flow constraint factorizes through features.
- That structure is exactly what the next slide exploits.

From the primal program to feature-level constraints

- P²IL [39] starts from the primal IL formulation.
- Under a linear MDP, both the reward and the dynamics depend on λ only through $\Phi^\top \lambda$.
- Introduce the feature-level occupancy $\rho = \Phi^\top \lambda$ to separate tabular and feature variables.

$$\begin{aligned} \max_{\lambda \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{A}|}} \min_{w \in \mathcal{W}} \langle \lambda - \lambda_{\pi_E}, \Phi w \rangle \\ \text{s.t. } E^\top \lambda = (1 - \gamma)\mu + \gamma M^\top \Phi^\top \lambda \end{aligned}$$

$\Downarrow$

$$\begin{aligned} \max_{\rho \in \Delta^m, \lambda \in \mathbb{R}^{\mathcal{S} \times \mathcal{A}}} \min_{w \in \mathcal{W}} \langle \rho - \Phi^\top \lambda_{\pi_E}^\mu, w \rangle \\ \text{s.t. } E^\top \lambda - \gamma M^\top \rho = (1 - \gamma)\mu, \\ \Phi^\top \lambda = \rho \end{aligned}$$

Remarks:

- The first constraint keeps discounted flow conservation in the MDP.
- The second constraint ties the tabular occupancy λ to its feature statistics ρ .
- This feature-level formulation is the starting point for the proximal algorithm.

Why the proximal step helps

- The feature-level formulation is still a hard max problem over Bellman errors.
- Proximal point regularizes the next iterate around the previous policy and occupancy.
- This turns the critic step into a smoothed log-sum-exp objective [39].

Logistic Bellman objective

$$\mathcal{G}_k(w, \theta) \triangleq -\frac{1}{\eta} \log \sum_{i=1}^m (\Phi^\top \lambda_{\pi_{k-1}})(i) e^{-\eta \delta_{w, \theta}^k(i)} + (1 - \gamma) \langle \mu, V_\theta^k \rangle - \langle \Phi^\top \lambda_{\pi_E}, w \rangle,$$
$$\delta_{w, \theta}^k \triangleq w + \gamma M V_\theta^k - \theta, \quad V_\theta^k(s) \triangleq -\frac{1}{\alpha} \log \sum_a \pi_{k-1}(a|s) e^{-\alpha Q_\theta(s, a)}, \quad Q_\theta = \Phi \theta.$$

Remarks:

- The log-sum-exp term is a smooth surrogate for the hard max Bellman error.
- We optimize cost and Q jointly, instead of alternating nested updates.
- This yields the soft-min actor update used on the next slide.

P²IL: proximal policy updates

Proximal Point Imitation Learning: P²IL

Initialize π_0 as uniform distribution over $\mathcal{A}$

for $k = 1, \dots, K$ **do**

 // Policy evaluation

$$(w_k, \theta_k) \approx \arg \max_{w \in \mathcal{W}, \theta \in \Theta} \mathcal{G}_k(w, \theta)$$

 // Policy improvement

$$\pi_k(a|s) \propto \pi_{k-1}(a|s) e^{-\alpha Q_{\theta_k}(s, a)}$$

end for

- Policy evaluation approximately maximizes the smooth objective $\mathcal{G}_k(w, \theta)$.
- Policy improvement is an exponentiated update toward actions with smaller $Q_{\theta_k}(s, a)$.

Remarks:

- Each iteration has one critic step and one actor step.
- There is no separate nested adversarial cost update inside policy evaluation.

P²IL: what the guarantee says

- At outer iteration k , the critic step solves $\mathcal{G}_k$ only approximately.
- Define the per-iteration error by

$$\epsilon_k = \mathcal{G}_k(w_k^*, \theta_k^*) - \mathcal{G}_k(w_k, \theta_k).$$

Error propagation

Let $\hat{\pi}_K$ be the average iterate. Then, with probability at least $1 - \delta$,

$$d_{\mathcal{C}}(\lambda_{\hat{\pi}_K}, \lambda_{\pi_E}) \leq \frac{1}{K} \left(\log(m|\mathcal{A}|) + C \sum_k \sqrt{\epsilon_k} + \sum_k \epsilon_k \right).$$

- How to read it:**
- The factor $\frac{1}{K}$ is the gain from the outer proximal loop.
 - The two sums accumulate the whole error sequence $(\epsilon_1, \dots, \epsilon_K)$.
 - If each ϵ_k is small, the occupancy mismatch decreases with K

$$\epsilon_k \leq \mathcal{O} \left(\frac{\max\{\eta, 1\}m}{\beta \sqrt{T}} \right) \implies K = \Omega(\epsilon^{-1}), T = \Omega(\epsilon^{-4}), \text{ total } \mathcal{O}(\epsilon^{-5}).$$

P²IL: what works and what does not

- What works: no inner RL loop, only a smooth stochastic optimization problem.
- What does not: the gradients of $\mathcal{G}_k$ must be estimated from exploratory samples.
- In REPS, [28] required the following full-support assumption.

Exploration assumption

We can sample state action pairs from an occupancy measure $\lambda_{\pi_0}(s, a) > 0 \quad \forall s, a \in \mathcal{S} \times \mathcal{A}$.

- In the linear MDP setting, this becomes a feature-space coverage assumption.

Positive definite covariance matrix

We can sample state action pairs from an occupancy measure λ_{π_0} such that.

$$\sigma_{\min} \left(\mathbb{E}_{s, a \sim \lambda_{\pi_0}} \phi(s, a) \phi(s, a)^\top \right) \geq \beta > 0.$$

ILARL: optimism replaces explicit coverage

- ILARL is introduced in [40].
- Key idea:** perform updates with *optimistic* estimates of the value function.
- This is in the spirit of OPPO [7], PPO/NPG with an optimistic evaluation step.
- In ILARL, this trades a hard coverage assumption for explicit, directed exploration.

Imitation Learning via Adversarial Reinforcement Learning: ILARL

For $k = 0, \dots, K - 1$, repeat:

$$r^{k+1} = \Pi_{\mathcal{R}} \left[r^k + \gamma(\lambda^{\pi_E} - \lambda^{\pi^k}) \right] \quad \text{reward player,}$$

$$Q^{k+1}(s, a) = r^{k+1}(s, a) + \gamma \phi(s, a)^T \theta^k + b^k(s, a) \quad \text{optimistic critic,}$$

$$\pi_{k+1}(a|s) \propto \pi_k(a|s) e^{\eta Q^{k+1}(s, a)} \quad \text{policy player.}$$

- Why it helps:**
- The reward update still pushes learner and expert occupancies together.
 - The bonus b^k makes uncertain actions look better, so the learner explores them.
 - This removes the old coverage assumption and improves the rate from $\mathcal{O}(\epsilon^{-5})$ to $\mathcal{O}(\epsilon^{-4})$.

Guarantees for ILARL

- The following guarantee is from [40].
- It cleanly separates MDP interaction complexity from expert-data complexity.

Theorem

After using $\tilde{\mathcal{O}}\left(\frac{\log|\mathcal{A}|d^3}{(1-\gamma)^8\epsilon^4}\right)$ state action pairs from the MDP and using $\tilde{\mathcal{O}}\left(\frac{2d\log(2d)}{(1-\gamma)^2\epsilon_E^2}\right)$ expert demonstrations ILARL outputs a policy which is at most $\epsilon + \epsilon_E$ -suboptimal, i.e.

$$\mathbb{E}[\langle \mu, V^{\pi^*} - V^{\pi^{\text{out}}} \rangle] \leq \epsilon + \epsilon_E.$$

How to read it: ◦ The first term counts learner interaction with the MDP.

- The second term counts expert demonstrations, through the expert estimation error ϵ_E .
- The theorem is model-free: the transition kernel need not be known.
- In linear MDPs with reward features only, the lower bound is $\tilde{\mathcal{O}}\left(\frac{d}{(1-\gamma)^3\epsilon^2}\right)$ [40].
- This leaves room to improve the current $\tilde{\mathcal{O}}(\epsilon^{-4})$ upper bound toward $\tilde{\mathcal{O}}(\epsilon^{-2})$.

IQ-Learn: learning Q directly from demonstrations

- IQ-Learn [14] is one of the most influential recent imitation learning algorithms.
- Key idea: learn a soft state-action value function directly from expert data.
- This keeps the objective simple in practice, and avoids an explicit inner min-max loop.

Worst-case imitation objective

$$\min_{\pi} d_{\mathcal{R}}(\pi, \pi_E) = \min_{\pi} \max_{r \in \mathcal{R}} \left(\rho_r(\pi_E) - \rho_r(\pi) \right) = \min_{\lambda \in \tilde{\mathcal{R}}} \max_{r \in \mathcal{R}} \langle \lambda_{\pi_E} - \lambda, r \rangle,$$

where $\rho_r(\pi) := (1 - \gamma) \langle \mu, V_r^{\pi} \rangle$.

IQ-Learn: a smoothed saddle formulation

- IQ-Learn can be viewed as a doubly smoothed version of the occupancy-measure saddle problem.
- Add an entropy-like regularizer on occupancies and a concave regularizer on rewards.

$$\min_{\lambda \in \mathcal{F}} \max_r \langle \lambda_{\pi_E} - \lambda, r \rangle + \frac{1}{\chi} \psi(r) + \frac{1}{\eta} H(\lambda, \lambda_0),$$

where H is the relative conditional entropy defined as

$$H(\lambda, \lambda_0) := \sum_{x,a} \lambda(x,a) \log \frac{\lambda(x,a) \sum_a \lambda_{\pi_0}(x,a)}{\lambda_{\pi_0}(x,a) \sum_a \lambda(x,a)}.$$

- The reward regularizer is chosen as $\psi(r) = \langle \lambda_{\pi_E}, r - \phi(r) \rangle$, where ϕ is convex and non-increasing.

IQ-Learn: equivalent unconstrained problem

IQ-Learn program over Q -functions

Replacing the optimal policy $\pi_Q(a|s) \propto \exp(Q(s, a))$ and let $V_Q(s) = \log \sum_{a \in \mathcal{A}} \exp(Q(s, a))$, we obtain an unconstrained problem.

$$\tilde{Q} \approx \arg \max_Q (1 - \gamma) \langle \mu, V_Q \rangle - \langle \lambda_{\pi_E}, \phi(Q - \gamma PV_Q) \rangle$$

Remarks:

- The objective is close in spirit to REPS, but optimized directly over Q .
- It solves the feature-matching problem without explicit min-max updates.
- The unconstrained derivation is not free, it requires assumptions on ψ .
- Strong empirical performance, but no general convergence guarantee.

Comparison between IQ-Learn and P²IL

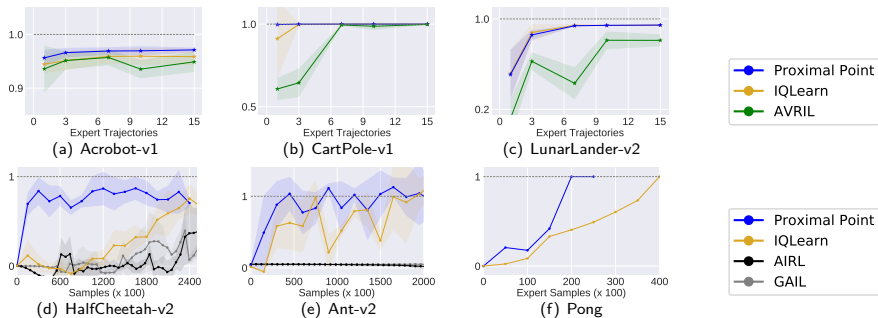


Figure: Offline IL and continuous control experiments

Deep IL still needs exploration

- ILARL made clear that exploration matters once we leave pure behavioral cloning.
- Its optimism bonus is explicit and linear: $b(s, a) = \sqrt{\phi(s, a)^T \Lambda \phi(s, a)}$.
- That works with linear features, but not naturally with neural critics.
- IL-SOAR [42] replaces explicit bonuses with ensemble-based optimism.

IL-SOAR: optimism from critic ensembles

- The paper's practical message is modular: add optimistic critic ensembles to SAC-based IL [42].
- View it as a primal-dual loop that alternates between reward learning and policy improvement.
- Instead of one SAC critic [16], train an ensemble of critics.
- Critic disagreement acts as a confidence region, and hence as an exploration signal.

IL-SOAR

Require: Reward step size α , Expert dataset $\mathcal{D}_{\pi_E}$, Discount factor γ , Policy step size η .

Initialize π^1 as uniform distribution over $\mathcal{A}$.

Initialize empty replay buffer, i.e. $\mathcal{D}^0 = \{\}$

for $k = 1$ to K **do**

$\tau^k \leftarrow \text{COLLECTTRAJECTORY}(\pi^k)$

 Add τ^k to replay buffer, i.e. $\mathcal{D}^k = \mathcal{D}^{k-1} \cup \tau^k$.

$r^k \leftarrow \text{UPDATEREWARD}(\tau^{k-1}, \mathcal{D}_{\pi_E}, \mathcal{D}^k, \alpha)$

for $\ell = 1$ to L **do**

 Compute estimator Q_ℓ^k .

end for

$Q^k = \text{OPTIMISTICQ}(\{Q_\ell^k\}_{\ell=1}^L)$.

$\pi^{k+1}(a|s) = \text{POLICYUPDATE}(\eta, \{Q^\tau(s, a)\}_{\tau=1}^k)$

end for

IL-SOAR: what is known theoretically?

- Current guarantees are for the tabular case.

Theorem

Let us consider the SOAR algorithm of [42] run in a tabular MDP for $K = \tilde{\mathcal{O}}\left(\frac{|S|^4|\mathcal{A}|\log(1/\delta)}{(1-\gamma)^5\epsilon^2}\right)$ iterations and with an expert dataset of size $|\mathcal{D}_{\pi_E}| = \frac{|S|^2|\mathcal{A}|\log(|S||\mathcal{A}|/\delta)(\log(|S|)+2)^2}{\epsilon^2(1-\gamma)^2}$. Then, it holds that the policy output by SOAR $\hat{\pi}_K$ satisfies $\left\langle \mu, V_{c_{\text{true}}}^{\pi_E} - V_{r_{\text{true}}}^{\hat{\pi}_K} \right\rangle \leq \epsilon$ with probability at least $1 - 5\delta$.

Remark: ○ In state-only imitation, the $\mathcal{O}(\epsilon^{-2})$ dependence cannot be improved.

How IL-SOAR implements optimism with neural critics

- With neural critics, SOAR uses disagreement as a practical proxy for epistemic uncertainty [42].
- For neural critics, optimism is estimated from disagreement across an ensemble.
- For each sampled state-action pair, use the mean value plus a clipped standard deviation.

Optimistic critic from an ensemble

Require: Replay buffer $\mathcal{D}$, Estimators $\{Q_\ell\}_{\ell=1}^L$, maximum standard deviation σ .

- 1: $\{s_i\}_{i=1}^N \leftarrow$ sample observations from $\mathcal{D}$
- 2: $a_i \leftarrow \pi(s_i)$
- 3: $\bar{Q}(s_i, a_i) = \frac{1}{L} \sum_{\ell=1}^L Q_\ell(s_i, a_i)$
- 4: $\text{std-Q}(s_i, a_i) = \sqrt{\frac{1}{L} \sum_{\ell=1}^L (Q_\ell(s_i, a_i) - \bar{Q}(s_i, a_i))^2}$
- 5: $\overline{\text{std-Q}}(s_i, a_i) \leftarrow \text{Clip}(\text{std-Q}(s_i, a_i), 0, \sigma)$.
- 6: $\overline{Q}(s_i, a_i) = \bar{Q}(s_i, a_i) + \overline{\text{std-Q}}(s_i, a_i)$
- 7: **Return:** $\overline{Q}(s_i, a_i)$ for all $i = 1, \dots, N$.

IL-SOAR improves SAC-based imitators

- Across MuJoCo tasks, adding SOAR improves both sample efficiency and final return [42].
- IL-SOAR is a general exploration template, not a single standalone baseline.
- Read each plot as: a SAC-based IL method versus the same method with optimistic critics.
- The gains are consistent across CSIL, RKL, and ML-IRL [44, 26, 46].

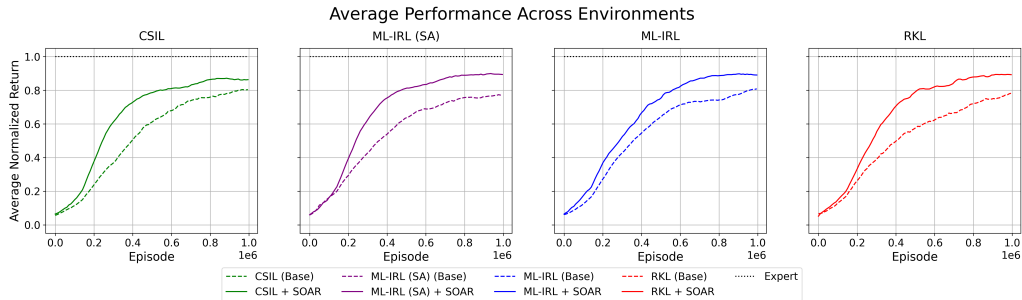


Figure: Average normalized return across 4 MuJoCo environments with 16 expert trajectories.

Beyond actions: imitating intent

Standard imitation learning

- Copy the *actions* performed by the expert.
- Usually does not model why those actions are good.



Figure: Robot imitation

Human imitation learning

- Infer the *intent* behind the behavior.
- New situations may call for different actions.



Figure: Human imitation

Inverse reinforcement learning: infer a reward

IRL Objective

Find reward function $r(\cdot, \cdot) : \mathcal{S} \times \mathcal{A} \rightarrow [-1, 1]$ that explains the expert's behavior:

$$\pi_E \in \arg \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \mu, \pi \right].$$

Inverse reinforcement learning: infer a reward

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$$\pi_E \in \arg \max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 \sim \mu, \pi \right].$$

Namely, it holds that

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 \sim \mu, \pi_E \right] \geq \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) | s_0 \sim \mu, \pi \right], \forall \pi \in \Pi.$$

- Remarks:**
- Classical formulations go back to [35, 25].
 - Assume the expert is approximately optimal for some unknown reward r_{true} .
 - Unlike BC, IRL uses the MDP structure, not only action labels.
 - The payoff is better reasoning about intent, but the optimization is harder.

RL vs IRL: what changes?

	IRL	RL
Input	Expert Demonstrations	Reward Function
Output	Optimal policy Reward function	Optimal Policy

- RL recovers a nearly optimal behavior from reward functions.
- IRL starts from demonstrations and asks which reward would make them optimal.

Why IRL is harder than it sounds

- First obstacle: behavior alone does not pin down a unique reward [25].

Theorem (Potential-based reward shaping)

The potential-based shaping result of [25] says that an expert policy π_E optimal in the MDP $\mathcal{M}$ with reward r is also optimal in the same MDP with reward function $\hat{r}$ given by

$$\hat{r}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot | s, a)} [\Phi(s')] - \Phi(s),$$

where $\Phi : \mathcal{S} \rightarrow \mathbb{R}$ is called potential function.

Remarks:

- So IRL is underdetermined in a single environment.
- Evaluating a candidate reward typically requires solving a planning problem.
- With finite trajectories and imperfect experts, the constraints must be estimated and relaxed.

Identifiability in inverse reinforcement learning

- One way to break reward ambiguity is to observe behavior under different dynamics [32].

Theorem (Two-environment identifiability)

Consider two Markov decision problems on the same set of states and actions, but with different transition matrices P^1, P^2 and discount factors γ_1, γ_2 . Suppose that we observe two experts acting each in one of these environments, optimally with respect to the same reward function, in the sense that their policies maximize the entropy regularized reward in their respective environments. Then, the reward function can be recovered up to the addition of a constant if and only if

$$\text{rank} \begin{pmatrix} I - \gamma_1 P_{a_1}^1 & -(I - \gamma_2 P_{a_1}^2) \\ \vdots & \vdots \\ I - \gamma_1 P_{a_{|\mathcal{A}|}}^1 & -(I - \gamma_2 P_{a_{|\mathcal{A}|}}^2) \end{pmatrix} = 2|\mathcal{S}| - 1. \quad (9)$$

Remark:

- A single environment is not enough because of shaping ambiguity.
- Two environments can pin down the reward up to an additive constant.
- This matters when we want to predict behavior under changed dynamics but shared intent.

Takeaways from imitation learning

Method	Reward learning	Access to environment	Interactive demonstrations	Pre-collected demonstrations
Behavioural Cloning	NO	NO	NO	YES
Online IL	NO	YES	YES	MAYBE
Inverse RL	YES	YES	NO	YES
Adversarial IL	MAYBE	YES	NO	YES
Non-adversarial IL	MAYBE	YES	NO	YES

- Remarks:**
- BC is the simplest option, but suffers from covariate shift and cascading errors.
 - Interactive IL corrects the distribution mismatch, but needs expensive expert queries.
 - IRL reasons about intent through rewards, at the price of harder optimization and scalability.
 - Modern deep IL revolves around two themes: stable optimization and effective exploration.

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Supplementary Material

Proof Sketch

- Recall the advantage defined as $A^{\hat{\pi}}(s, a) = Q^{\hat{\pi}}(s, a) - V^{\hat{\pi}}(s)$ and notice that $\mathbb{E}_{a \sim \hat{\pi}(\cdot|s)} A^{\hat{\pi}}(s, a) = 0, \quad \forall s.$
- We will use also that $A^{\hat{\pi}}(s, a) \leq \frac{1}{1-\gamma}$ if $\max_{s,a} |r(s, a)| \leq 1.$

Proof.

- Based on performance difference lemma [21], we have

$$\begin{aligned} V^{\pi_E} - V^{\hat{\pi}} &= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \lambda^{\pi_E}, a \sim \pi_E(\cdot|s)} A^{\hat{\pi}}(s, a) \\ &= \frac{1}{1-\gamma} \left[\mathbb{E}_{s \sim \lambda^{\pi_E}, a \sim \pi_E(\cdot|s)} A^{\hat{\pi}}(s, a) - \mathbb{E}_{s \sim \lambda^{\pi_E}, a \sim \hat{\pi}(\cdot|s)} A^{\hat{\pi}}(s, a) \right] \\ &\leq \frac{1}{(1-\gamma)^2} \mathbb{E}_{s \sim \lambda^{\pi_E}} \|\hat{\pi}(\cdot|s) - \pi_E(\cdot|s)\|_1. \end{aligned}$$

□

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- MLE guarantee [4] is given by

$$\mathbb{E}_{s \sim \lambda^{\pi_E}} \|\hat{\pi} - \pi_E\|_{TV}^2 \leq \frac{\log(|\Pi|/\delta)}{|\mathcal{D}|}.$$

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$$\mathbb{E}_{s \sim \lambda^{\pi_E}} \|\hat{\pi} - \pi_E\|_{TV}^2 \leq \frac{\log(|\Pi|/\delta)}{|\mathcal{D}|}.$$

- Then the result follows from Jensen's inequality and that $\|\cdot\|_{TV} = \frac{1}{2} \|\cdot\|_1.$

□

★ Hoeffding's Lemma

- We use Hoeffding's lemma from [19].

Theorem (Hoeffding's Lemma)

Let X be a random variable such that $\mathbb{E}(X) = 0$ and $X \in [a, b]$ almost surely. Then for any $s \in \mathbb{R}$, it holds that

$$\mathbb{E}(e^{sX}) \leq e^{\frac{s^2(b-a)^2}{8}}.$$

Generative adversarial imitation learning (GAIL): A primal dual perspective

- In Maximum Causal Entropy IRL [48], we need to solve an RL problem for every reward update.
- This is a major computation bottleneck.
- We can develop a more efficient method if we use alternating updates.

Derivation: ◦ Let us consider a reward linear in some features, that is $r(s, a) = \langle \phi(s, a), w \rangle$. ◦ We will follow the same steps from [17]

GAIL objective

Let $h : \mathbb{R}^{|S||A|} \rightarrow \mathbb{R}$ be a convex function that serves as reward regularizer. GAIL solves the following minimax problem:

$$\min_r \max_{\pi \in \Pi} \quad \beta h(r) + \mathbb{E}_{s,a \sim \lambda_\mu^\pi} [r(s, a)] - \mathbb{E}_{s,a \sim \lambda_\mu^{\pi_E}} [r(s, a)] + \alpha \mathbb{E}_{s,a \sim \lambda_\mu^\pi} [-\log \pi(a|s)]$$

- Use Fenchel conjugation, we can obtain

$$\max_{\pi \in \Pi} \quad -h^*(\Phi^T \lambda_\mu^{\pi_E} - \Phi^T \lambda_\mu^\pi) + \alpha \mathbb{E}_{s,a \sim \lambda_\mu^\pi} [-\log \pi(a|s)].$$

- Important result: If f is α strongly convex then the convex conjugate f^* is $1/\alpha$ -smooth [5].

An important choice for the regularizer h .

- Choosing h as

$$h(r) = \begin{cases} \mathbb{E}_{s,a \sim \lambda_\mu^{\pi_E}} [g(r(s,a))], & \text{if } r(s,a) < 0; \\ \infty, & \text{otherwise.} \end{cases}$$

with $g(x) = -x - \log(1 - e^x)$.

- The Fenchel conjugate of h is given by:

$$h^*(\lambda_\mu^{\pi_E} - \lambda_\mu^\pi) = \max_{D \in [0,1]} \mathbb{E}_{s,a \sim \lambda_\mu^\pi} [\log D(s,a)] + \mathbb{E}_{s,a \sim \lambda_\mu^{\pi_E}} [\log(1 - D(s,a))]$$

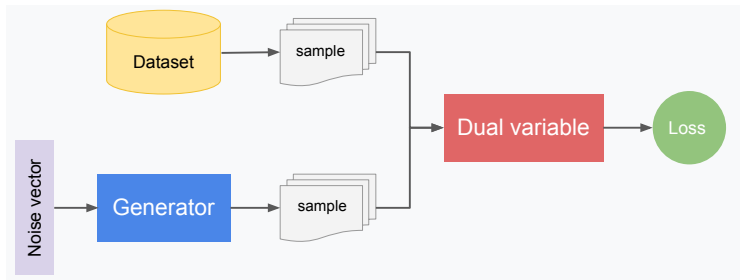
that is widely known as the (vanilla) GAN loss.

- Therefore, we can learn a policy from demonstrations solving the following saddle point problem:

$$\min_{\pi \in \Pi} \max_{D \in [0,1]} \mathbb{E}_{s,a \sim \lambda_\mu^\pi} [\log D(s,a)] + \mathbb{E}_{s,a \sim \lambda_\mu^{\pi_E}} [\log(1 - D(s,a))] - \alpha \mathbb{E}_{s,a \sim \lambda_\mu^\pi} [-\log \pi(a|s)].$$

Generative Adversarial Network (GANs)

- GAN [15] is framed as a min-max game between a generator and a discriminator.



- GAN: ($\Rightarrow$ minimizing the Jensen-Shannon divergence)

$$\min_{G_\phi} \max_{D_\theta} \mathbb{E}_{x \sim p_{\text{data}}} [\log D_\theta(x)] + \mathbb{E}_z [\log(1 - D_\theta(G_\phi(z)))]$$

- Wasserstein GAN: ($\Rightarrow$ minimizing the Wasserstein divergence)

$$\min_{G_\phi} \max_{f_\theta: 1\text{-Lipschitz}} \mathbb{E}_{x \sim p_{\text{data}}} [f_\theta(x)] - \mathbb{E}_z [f_\theta(G_\phi(z))]$$

Generative Adversarial Networks (GANs)



**2014
GAN**



**2018
GAN**

Generative Adversarial Imitation Learning (GAIL)

- GAIL [18] aims to solve the min-max game for learning the policy given an expert policy π_E .

$$\min_{\theta} \max_{\phi} \mathbb{E}_{s,a \sim \lambda^{\pi_{\theta}}} [\log(D_{\phi}(s,a))] + \mathbb{E}_{s,a \sim \lambda_{\mu}^{\pi_E}} [\log(1 - D_{\phi}(s,a))] - \alpha H(\pi_{\theta}).$$

- Remarks:**
- We assume a differentiable parametrized policy π_{θ} .
 - The discriminator tries to separate the data generated from learned policy from expert data.
 - Equivalent to minimize the Jensen-Shannon divergence between the state-action distributions of the expert policy and the learned policy.
 - Unlike Max-Entropy IRL, does not require expensive RL subroutines to learn the reward.
 - GAIL can be adapted to use features only datasets [38].

Numerical performance

- Results from GAIL [18].

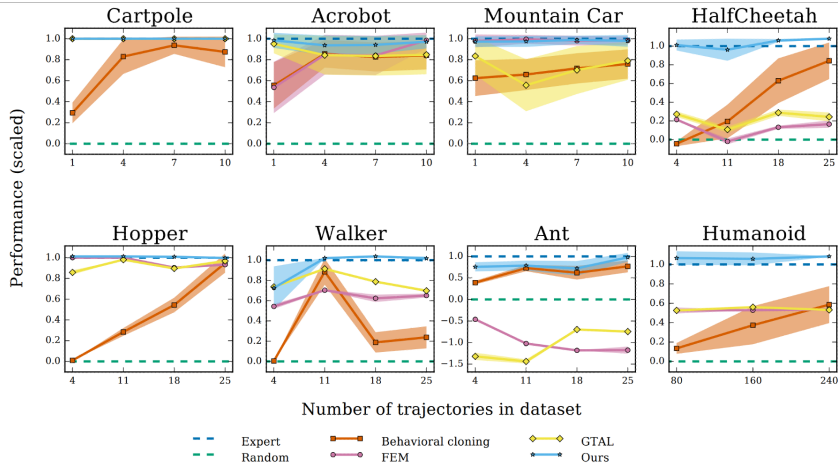


Figure: Performance of learned policies among GAIL, Behavior Cloning (BC), Feature Expectation Matching (FEM), and Game-theoretic Apprenticeship Learning (GATL)

Feature-based reward

Theorem

Assumption Let $\phi : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}^d$ be a feature mapping. Assume linear true reward function, i.e.,

$$r_{true} \in \{r \mid r(s, a) = w^\top \phi(s, a), \text{ where } w \in \mathbb{R}^d \text{ and } \|w\|_2 \leq 1\}.$$

◦ The expected total reward when $r(s, a) = w^\top \phi(s, a)$ can then be expressed as:

$$J_r(\pi) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \middle| \pi \right] = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t w^\top \phi(s_t, a_t) \middle| \pi \right] = w^\top \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, a_t) \middle| \pi \right] = w^\top \rho_\phi(\pi),$$

where $\rho_\phi(\pi) \in \mathbb{R}^d$ is the feature expectation vector of policy π .

Goal

Find $w \in \mathbb{R}^d$ such that

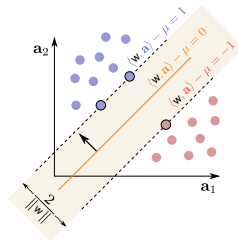
$$\underbrace{w^\top \rho_\phi(\pi_E)}_{=J_w(\pi_E)} \geq \underbrace{w^\top \rho_\phi(\pi)}_{=J_w(\pi)}, \quad \forall \pi \in \Pi.$$

Feature-based reward (cont'd)

Goal

Find $w \in \mathbb{R}^d$ such that

$$\underbrace{w^\top \rho_\phi(\pi_E)}_{=J_w(\pi_E)} \geq \underbrace{w^\top \rho_\phi(\pi)}_{=J_w(\pi)}, \quad \forall \pi \in \Pi.$$



Remark:

- Note that $\rho_\phi(\pi)$ can be readily estimated from sampled trajectories.
- By Hoeffding's Lemma [19] (see 13) we need $\mathcal{O}\left(\frac{d \log(\frac{1}{\delta})}{(1-\gamma)^2 \varepsilon^2}\right)$ expert trajectories to have an ε -small ℓ_∞ -error with probability at least $1 - \delta$.

Max margin IRL

- This max-margin IRL formulation is due to Ratliff et al. [31].
- The structured-prediction variant below is due to Taskar et al. [37].

Standard max-margin formulation

We want to maximize the *margin*, i.e the separation distance between the expert and other policies, this yields

$$\begin{aligned} \min_w \quad & \|w\|_2^2 \\ \text{s.t.} \quad & w^\top \rho_\phi(\pi_E) \geq w^\top \rho_\phi(\pi) + 1, \quad \text{for all } \pi \end{aligned}$$

Structured prediction max margin

We add flexibility by specifying the margin as a function of the policies, i.e., $m(\pi_E, \pi)$, this yields

$$\begin{aligned} \min_w \quad & \|w\|_2^2 \\ \text{s.t.} \quad & w^\top \rho_\phi(\pi_E) \geq w^\top \rho_\phi(\pi) + m(\pi_E, \pi), \quad \text{for all } \pi \end{aligned}$$

Remarks: ◦ We want to make $J_w(\pi_E)$ larger than any other $J_w(\pi)$ by a margin $m(\pi_E, \pi)$.

◦ Margin should be larger for policies that are very different from π_E .

Max margin IRL (cont')

- Continuation of the max-margin formulation of Ratliff et al. [31].

Structured prediction max-margin with slack variables

We relax the problem by allowing the constraints to be violated by introducing slack variables $\xi \geq 0$, this yields

$$\begin{aligned} \min_{w, \xi} \quad & \|w\|_2^2 + C\xi \\ \text{s.t.} \quad & w^\top \rho_\phi(\pi_E) \geq w^\top \rho_\phi(\pi) + m(\pi_E, \pi) - \xi, \quad \text{for all } \pi \end{aligned}$$

- Remarks:**
- The slack variable $\xi \geq 0$ are introduced to allow the constraints to be violated.
 - Resolved: access to π_E , reward ambiguity, expert suboptimality.
 - One challenge remains: very large number of constraints.
 - Assuming access to an RL subroutine, it can be solved, e.g., by constraint generation.